

Zeroth Review

CAREER CRAFT

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GUIDE

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Problem Statement

- **Industry Challenge:** Intense competition in campus placements demands technical, aptitude, and soft skills proficiency.
- **Traditional Preparation Issues:** Manual, time-consuming, inconsistent feedback, limited domain-specific resources.
- **Current Platform Limitations:** Costly, rigid, lack real-time analytics and AI-driven insights.
- **Need for Smart Automation:** Adaptive learning, real-time notifications, and skill gap analysis for tailored guidance.

Solution Approach

- AI-driven mock interviews with speech-to-text and feedback.
- NLP-based resume and cover letter generation with ATS optimisation.
- Adaptive technical and aptitude quizzes with analytics.
- Soft skills and confidence tracking using sentiment and speech analysis.
- Personalised learning roadmaps and skill gap analysis.
- Real-time notifications and performance tracking.
- Deployment on local or free cloud platforms using open-source AI/ML models for cost efficiency.

Modules

- **AI Mock Interview Simulator:**

AI-driven Mock interview with speech-to-text analysis and feedback on clarity/confidence.

- **Resume Builder:**

NLP-based resume builder job matching, and ATS optimization.

- **Technical Quiz Platform:**

Adaptive technical quizzes with analytics (Chart.js).

- **Student Performance Analyzer:**

ML-based tracking across quizzes, interviews, and skills of the students and overall analysis.

Modules

- **Company Knowledge Provider:**
Curated insights on company hiring processes using web scraping and NLP.
- **Aptitude Test Simulator:**
Adaptive tests for math, logical reasoning, and verbal skills.
- **Coding Challenge Platform:**
Timed coding problems with real-time feedback.
- **Group Discussion Simulator:**
AI-driven simulation with feedback on participation and clarity.
- **Typing Speed Improver:**
Gamified, ML-based adaptive coding exercises.

Modules

- **Soft Skills Heatmap:**

To develop Language proficiency, communication, time-management and leadership skills through tasks.

- **Skill Gap Analyzer:**

Compares user skills with job requirements, suggests resources.

- **Domain Selection and Preparation:**

Tailored quizzes, roadmaps, and questions for user-selected domains.

- **Real-Time Notification Provider:**

Socket.IO for alerts on Q&A, quiz updates, and company news.

Technology Stack

Front End

- React.js, JavaScript, Tailwind CSS, Redux Toolkit, Chart.js.

Back End

- Node.js, Express.js, MongoDB.

Languages

- JavaScript, Python.

Notifications

- Nodemailer (email).

Security/Privacy

- JWT for authentication, bcrypt for encrypted data, NLP-based moderation.

Thank you